

ABSTRACT

Smart home automation has become an essential part of modern living, enabling users to control home appliances intelligently and remotely. This mini project presents a clap sensor and app-controlled home automation system using the ESP32 microcontroller and the Blynk IoT platform. The system allows a relay-connected light to be controlled in two ways: by detecting the number of hand claps through a sound sensor, and by toggling a virtual button in the Blynk mobile application over Wi-Fi. A single clap turns the light OFF, and two consecutive claps turn it ON. Both control methods are synchronized using a shared state variable, ensuring consistent operation regardless of the input source. Edge-detection and debounce logic are implemented in firmware to accurately count claps without blocking the Wi-Fi communication loop. The proposed solution is low-cost, non-contact for the clap input, and demonstrates the potential of integrating IoT connectivity with physical sensor inputs for smart home applications.

KEY WORDS:

Smart Home Automation, ESP32, Blynk, Relay Control, Clap Detection, Sound Sensor, IoT, Wi-Fi Control, Debounce, Edge Detection, Embedded Systems, Arduino IDE.

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1. INTRODUCTION

Home automation refers to the automatic and manual control of household devices and appliances using technology. In recent years, the Internet of Things (IoT) has enabled affordable, connected, and intelligent home control systems that can be managed remotely from smartphones. At the same time, sensor-based inputs such as sound detection provide a hands-free, intuitive way for users to interact with their homes.

This mini project focuses on developing a smart home automation system that controls a relay-connected light using two input methods: a clap sensor and a mobile application. The ESP32 microcontroller acts as the central processing unit, handling Wi-Fi connectivity through the Blynk IoT platform and processing digital signals from a sound/clap sensor.

The system detects the number of claps within a defined time window and takes action accordingly — one clap turns the light OFF and two claps turn it ON. Simultaneously, the Blynk app button provides remote control over Wi-Fi. Both inputs share a common state variable, ensuring that the relay and app button always reflect the same light status regardless of which input was last used.

This project demonstrates how simple hardware components can be combined with modern IoT platforms to create practical and user-friendly smart home solutions suitable for everyday use.

1.1 OBJECTIVES

- Control a home appliance (light) using clap sensor input.
- Enable remote control of the appliance through the Blynk mobile application.
- Synchronize both input methods using a shared state variable.
- Implement reliable clap detection using edge detection and debounce logic.
- Demonstrate a low-cost IoT-based home automation prototype.
- Ensure the system remains responsive to Wi-Fi commands at all times without blocking delays.

2.LITERATURE REVIEW

Home automation systems have evolved significantly over the past decade, transitioning from simple timer-based controls to sophisticated IoT-enabled networks. Traditional automation relied on wired systems and dedicated controllers, which were expensive and complex to install. The advent of low-cost Wi-Fi microcontrollers such as the ESP8266 and ESP32 has democratized smart home development, making it accessible to students and hobbyists.

Clap-based or sound-activated control systems represent an early form of gesture-based human-machine interaction. Prior work has demonstrated their use in controlling lights, fans, and other appliances. However, challenges such as false triggering from ambient noise, debounce issues, and the inability to count multiple claps accurately have been well-documented in the literature. Modern approaches use edge detection, time-window-based counting, and morphological signal filtering to address these problems.

IoT platforms such as Blynk, Adafruit IO, and ThingSpeak have simplified the integration of microcontrollers with mobile applications. Blynk, in particular, provides a drag-and-drop interface for building control panels and supports real-time bidirectional communication between the device and the app. Several studies have used Blynk with ESP32 for relay-based home automation, validating its reliability for this class of applications.

The integration of multiple control modalities — physical sensors and remote app control — into a single synchronized system is a key challenge. Research has shown that using a shared state variable and centralizing all actuation logic into a single function is the most effective approach to maintaining consistency. This project builds upon these principles to create a robust and practical demonstration system.

3.WORKING PRINCIPLE

The Smart Home Automation system operates through two parallel control paths that are unified by a single shared state variable. The system block diagram shows the ESP32 at the centre, receiving inputs from the clap sensor and the Blynk app, and driving the relay module that controls the light.

1. Clap Sensor:

- Detects sound above a threshold and outputs a digital HIGH signal to the ESP32.
- Each physical clap produces one LOW-to-HIGH rising edge on the input pin.
- The firmware counts rising edges within a 1-second time window to determine the number of claps.
- One clap sets the state to OFF; two claps set the state to ON.

2. ESP32 Microcontroller:

- Central processing unit of the system.
- Connects to Wi-Fi and maintains a live connection with the Blynk server.
- Reads the clap sensor pin each loop iteration and detects rising edges using a previous-state variable.
- Runs the `applyState()` function to drive the relay and update the Blynk virtual pin simultaneously.

3. Blynk App (V0 Button):

- Provides a virtual ON/OFF button on the user's smartphone.
- Sends the button state (0 or 1) to the ESP32 via the Blynk server over Wi-Fi.
- The `BLYNK_WRITE(V0)` callback is invoked automatically when the button is pressed.
- The button display on the app is always kept in sync with the relay state.

4. Relay Module (Active-LOW):

- Controls the mains-powered light bulb.
- Driven LOW (relay ON, light ON) when state = 1.
- Driven HIGH (relay OFF, light OFF) when state = 0.

System Flow Summary:

1. Clap sensor detects sound and sends a digital pulse to ESP32.
2. ESP32 counts rising edges within a 1-second window.
3. After the window expires, the clap count is evaluated and `applyState()` is called.
4. `applyState()` writes to the relay pin and updates Blynk virtual pin V0.
5. Alternatively, pressing the Blynk button triggers `BLYNK_WRITE(V0)`, which also calls `applyState()`.
6. Both paths result in the same synchronized relay and app state.

Circuit Diagram

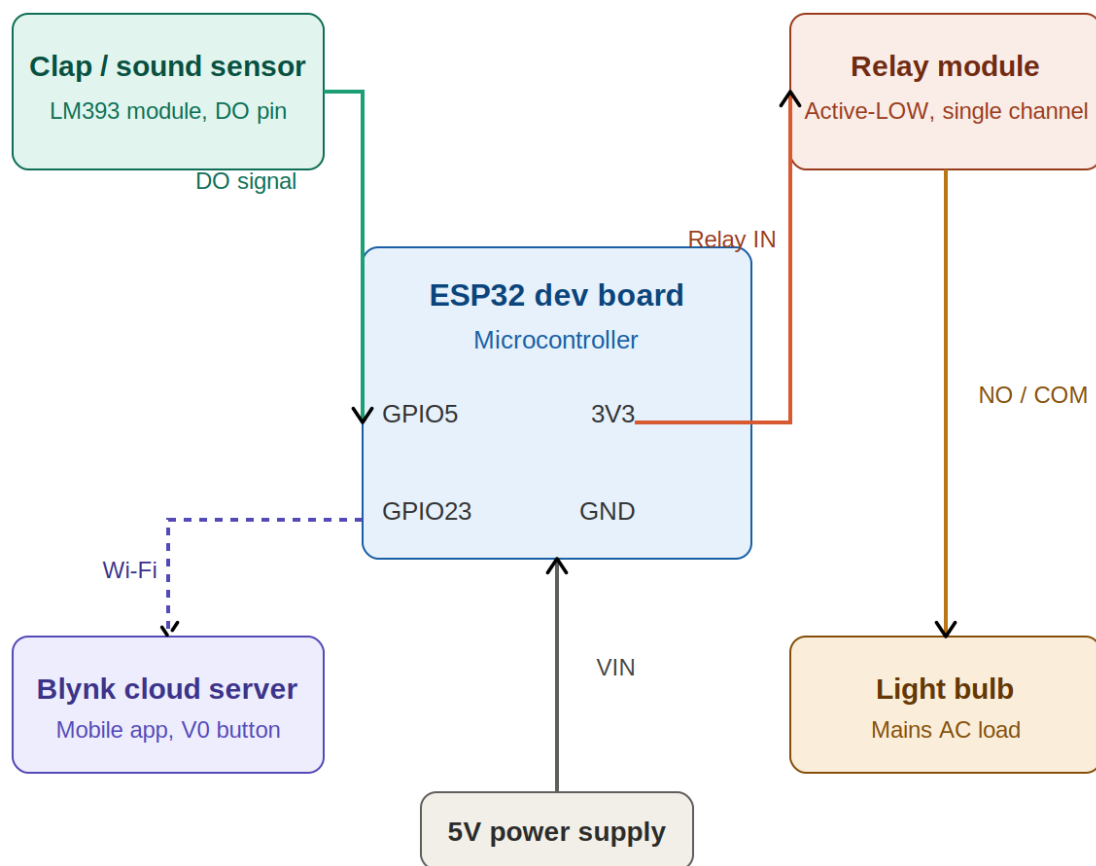


Fig: ESP32 based smart home automation - clap sensor, relay, bulb and Blynk app connectivity

4.COMPONENTS

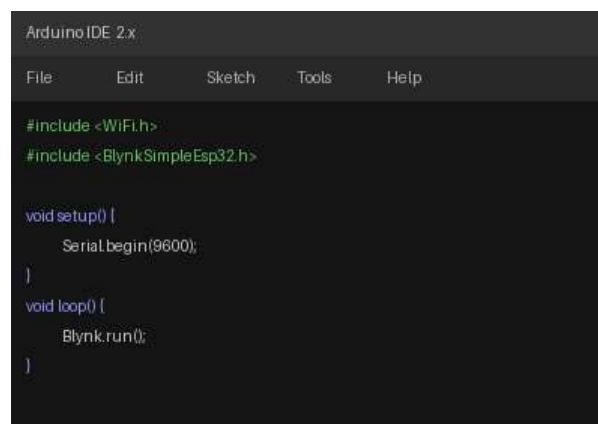
4.1 HARDWARE:

- ESP32 Development Board
- Clap / Sound Sensor Module
- Single-Channel Relay Module (Active-LOW)
- Light Bulb with holder
- Jumper Wires
- Breadboard
- USB Cable for programming
- Power Supply (5V)

4.2 SOFTWARE:

- Arduino IDE
- Blynk Mobile Application
- Blynk Library for ESP32

ARDUINO IDE:

A screenshot of the Arduino IDE 2.x interface. The window title is "Arduino IDE 2.x". The menu bar includes "File", "Edit", "Sketch", "Tools", and "Help". The code editor shows the following C++ code:

```
#include <WiFi.h>
#include <BlynkSimpleEsp32.h>

void setup() {
  Serial.begin(9600);
}

void loop() {
  Blynk.run();
}
```

The Arduino Integrated Development Environment (IDE) is an open-source software platform used to write, compile, and upload programs to microcontroller boards. It supports C and C++ programming and provides a large library ecosystem that simplifies hardware interfacing. In this project, the Arduino IDE is used to write the embedded firmware for the ESP32 board. The ESP32 board support package and the Blynk library are installed through the Arduino IDE's Library Manager and Board Manager. The Serial Monitor built into the IDE is also used during testing to observe clap detection events and debug the system in real time.

ESP32 DEVELOPMENT BOARD:



The ESP32 is a dual-core 32-bit microcontroller with built-in Wi-Fi and Bluetooth, making it ideal for IoT applications. It operates at up to 240 MHz and features multiple GPIO pins, ADC channels, and hardware PWM outputs. In this project, the ESP32 connects to the home Wi-Fi network and communicates with the Blynk server, while simultaneously monitoring the clap sensor pin and controlling the relay module. Its dual-core architecture ensures that Wi-Fi communication and sensor polling can occur efficiently without blocking each other.

CLAP / SOUND SENSOR MODULE:



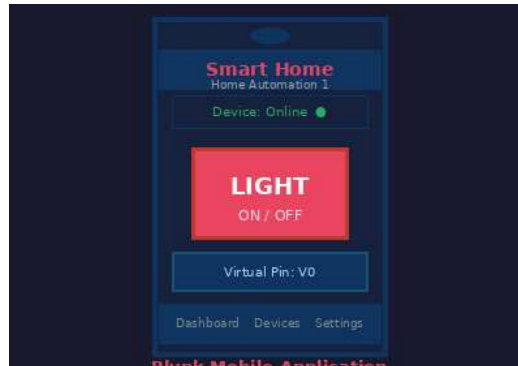
The clap sensor module contains a condenser microphone and a comparator circuit (typically LM393) that outputs a digital HIGH signal when the detected sound level exceeds an adjustable threshold. The sensitivity is tuned via an onboard potentiometer. In this project, the module is connected to a GPIO pin of the ESP32 and its digital output is polled each loop iteration. Rising-edge detection is used to identify individual claps, and a 1-second time window is used to count the number of consecutive claps before action is taken.

RELAY MODULE (ACTIVE-LOW):



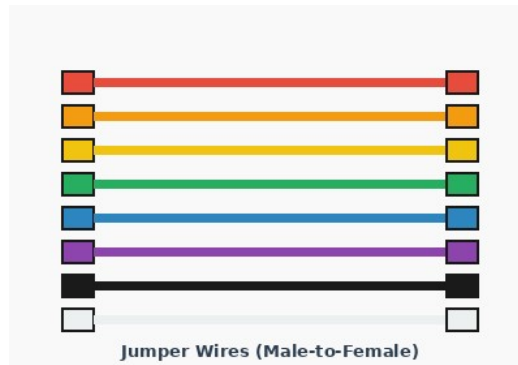
A relay is an electrically operated switch that allows a low-power microcontroller signal to control a high-power circuit such as a mains-connected light bulb. The dual-channel relay module used in this project is active-LOW, meaning the relay coil is energized (and the light turns ON) when the control pin is driven LOW by the ESP32. The module includes an optocoupler for electrical isolation between the microcontroller circuit and the mains circuit, ensuring safe operation. The initial state is set to HIGH (relay OFF) at power-on.

BLYNK APPLICATION:



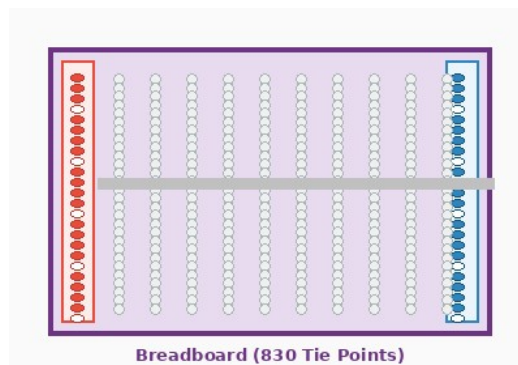
Blynk is an IoT platform that provides a mobile application (available on Android and iOS) for building custom dashboards to control and monitor microcontrollers over the Internet. A virtual button widget is placed on the Blynk dashboard and linked to virtual pin V0. When the button is pressed, the Blynk server sends the button value (0 or 1) to the ESP32, which processes it inside the `BLYNK_WRITE(V0)` callback. The app button state is kept synchronized with the relay using `Blynk.virtualWrite()`, ensuring the displayed state always reflects the actual light status.

JUMPER WIRES:



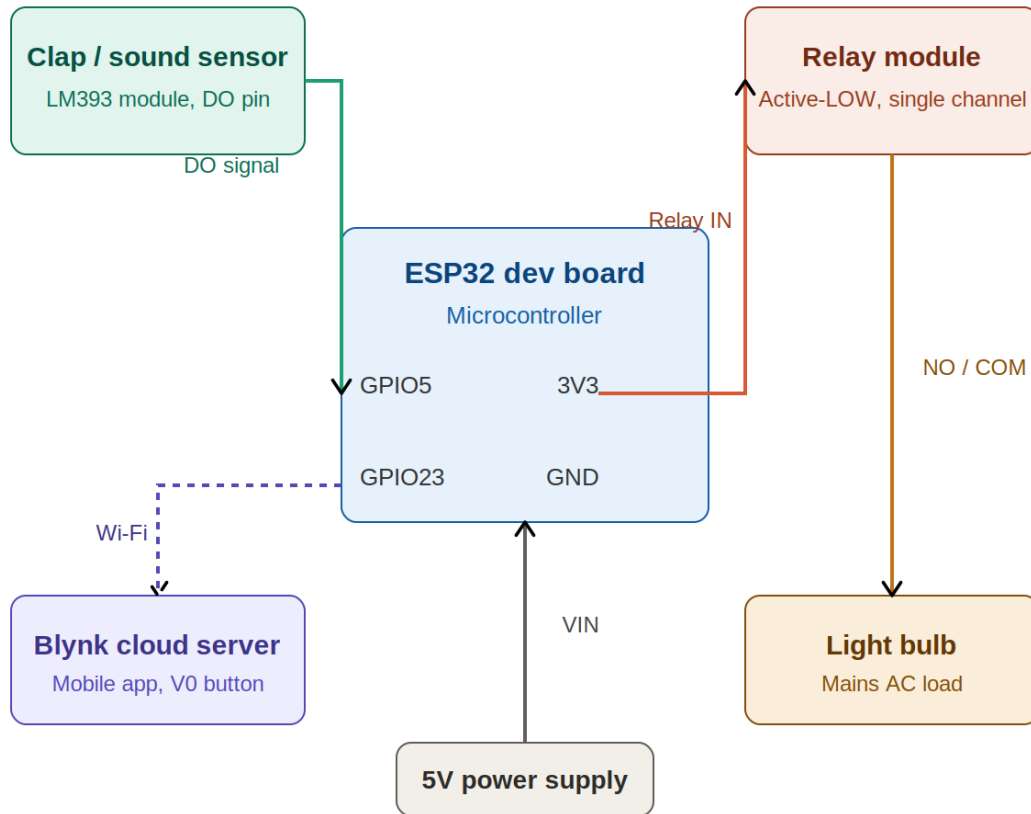
Jumper wires are used to make electrical connections between the ESP32 development board, the clap sensor module, and the relay module on the breadboard. Male-to-male and male-to-female jumper wires are used as required. They allow easy reconfiguration of the circuit without soldering, making the prototype flexible and easy to modify.

BREADBOARD:



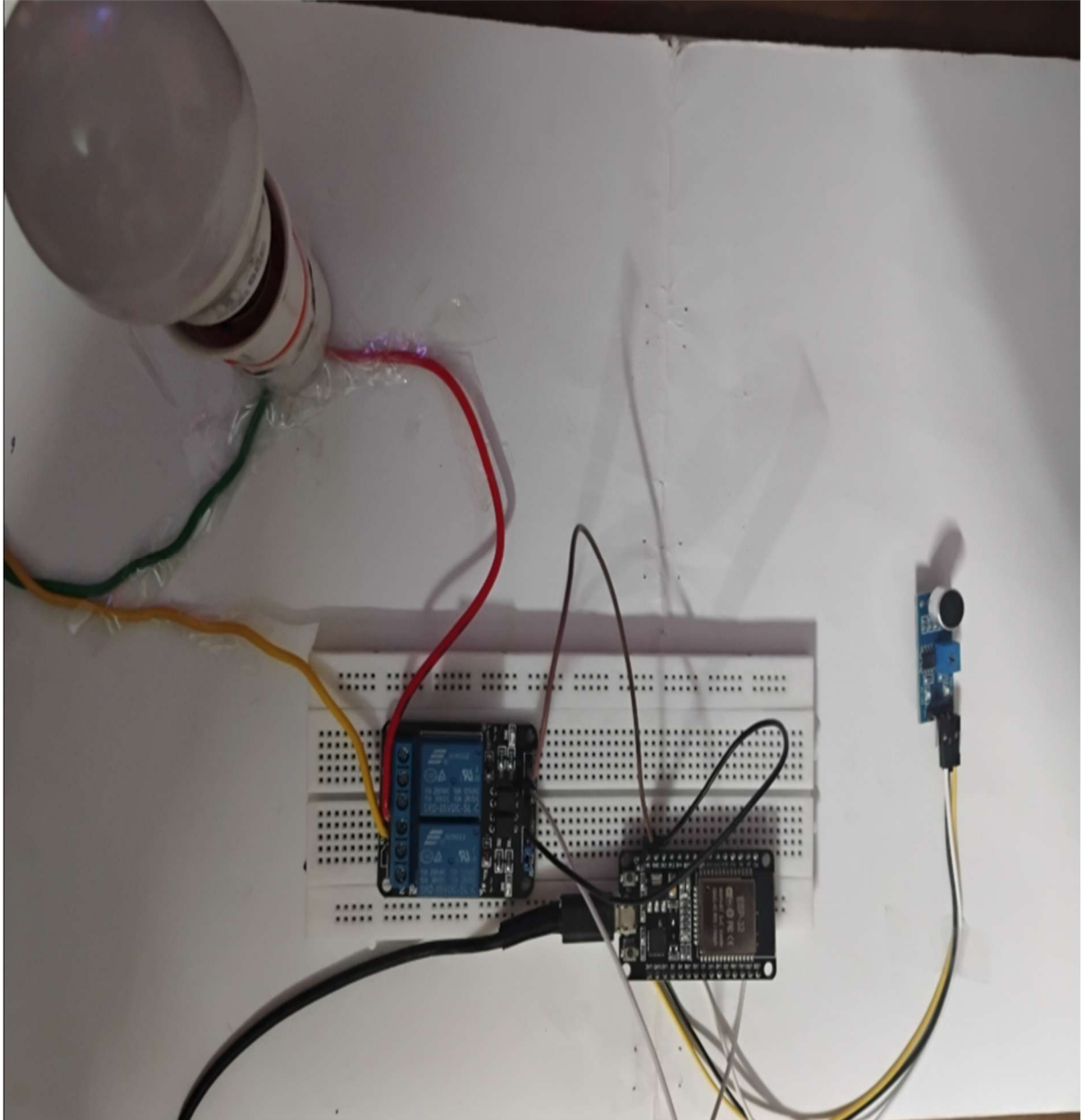
A breadboard is a solderless prototyping board used to build temporary electronic circuits. In this project, the breadboard is used to mount the ESP32 development board, clap sensor module, and relay module, and to route power and signal connections using jumper wires. It allows easy assembly, testing, and modification of the circuit without any permanent soldering.

CIRCUIT DIAGRAM



The ESP32 development board interfaces with the clap/sound sensor module on GPIO 5 and the relay module on GPIO 23. The sensor's digital output (DO) pin connects to GPIO 5, while power and ground are shared from the ESP32's 3V3 and GND pins. The relay module's IN pin connects to GPIO 23, with its NO/COM contacts switching the AC mains supply to the light bulb. The ESP32 connects wirelessly to the home Wi-Fi router, which links to the Blynk cloud server, enabling the Blynk mobile app to control virtual pin V0 in sync with the relay state.

HARDWARE:



5.C++ PROGRAM (EMBEDDED):

```
#define BLYNK_TEMPLATE_ID    "TMPL365K_p4wi"
#define BLYNK_TEMPLATE_NAME "Home Automation 1"
#define BLYNK_PRINT Serial

#include <WiFi.h>
#include <BlynkSimpleEsp32.h>

char auth[] = "LyTaj_205AQyiRAKMLzVV4MT0DyJwonq";
char ssid[] = "YourWiFiName";
char pass[] = "YourWiFiPassword";

const int RELAY_PIN = 23;
const int CLAP_PIN  = 5;

int  clapCount      = 0;
bool clapWindowOpen = false;
unsigned long clapWindowStart = 0;
unsigned long lastClapEdge    = 0;

const unsigned long DEBOUNCE_MS    = 200;
const unsigned long CLAP_WINDOW_MS = 1000;

int  relayState      = 0;
bool lastClapPinState = LOW;

// Helper: apply state to relay and sync Blynk
void applyState(int newState) {
    relayState = newState;
    digitalWrite(RELAY_PIN, relayState ? LOW : HIGH);
    Blynk.virtualWrite(V0, relayState);
}

// Blynk button on V0
BLYNK_WRITE(V0) {
    applyState(param.asInt());
}

void setup() {
    Serial.begin(9600);
    pinMode(RELAY_PIN, OUTPUT);
    pinMode(CLAP_PIN, INPUT);
    digitalWrite(RELAY_PIN, HIGH);
    Blynk.begin(auth, ssid, pass);
}

void loop() {
    Blynk.run();

    bool currentClapPin = digitalRead(CLAP_PIN);
```

```

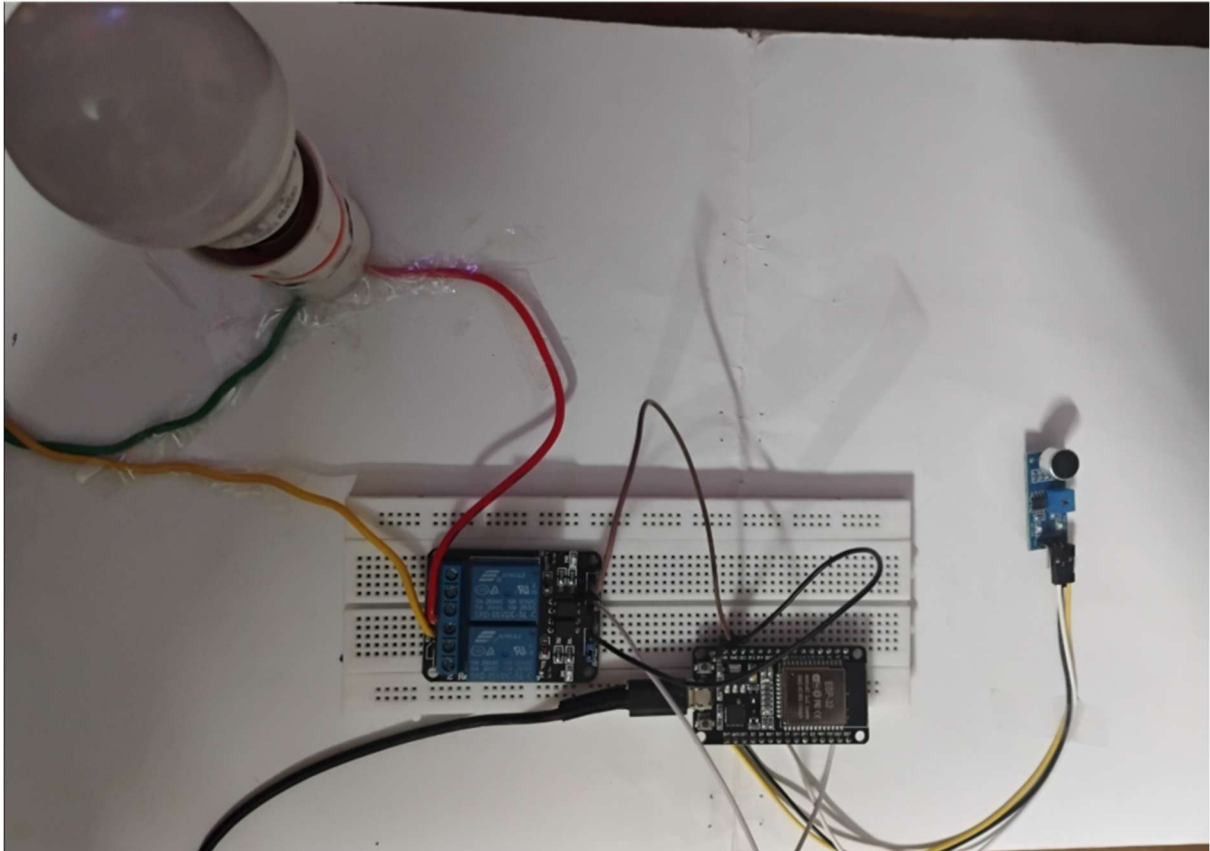
if (currentClapPin == HIGH && lastClapPinState == LOW) {
    unsigned long now = millis();
    if (now - lastClapEdge > DEBOUNCE_MS) {
        lastClapEdge = now;
        clapCount++;
        if (!clapWindowOpen) {
            clapWindowOpen = true;
            clapWindowStart = now;
        }
        Serial.print("Clap detected! Count: ");
        Serial.println(clapCount);
    }
}
lastClapPinState = currentClapPin;

if (clapWindowOpen && (millis() - clapWindowStart > CLAP_WINDOW_MS)) {
    if (clapCount == 1) {
        Serial.println("1 clap -> OFF");
        applyState(0);
    }
    else if (clapCount == 2) {
        Serial.println("2 claps -> ON");
        applyState(1);
    }
    else {
        Serial.print("Ignored (");
        Serial.print(clapCount);
        Serial.println(" claps)");
    }
    clapCount = 0;
    clapWindowOpen = false;
}
}

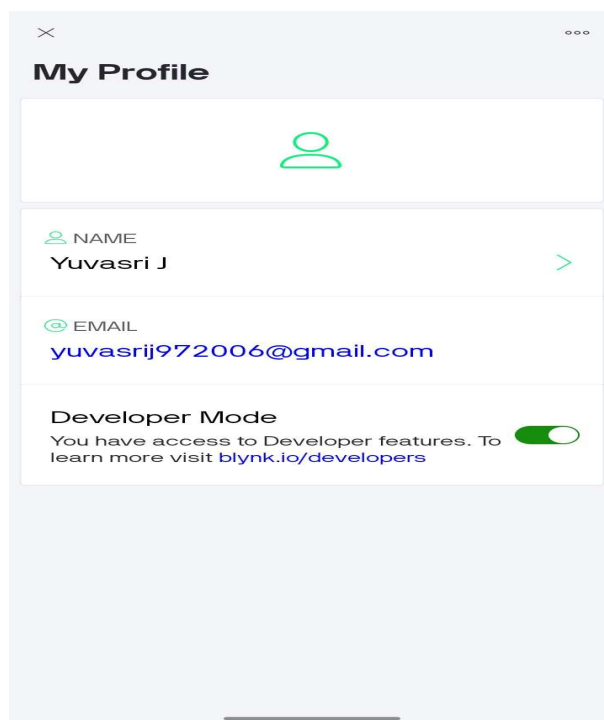
```

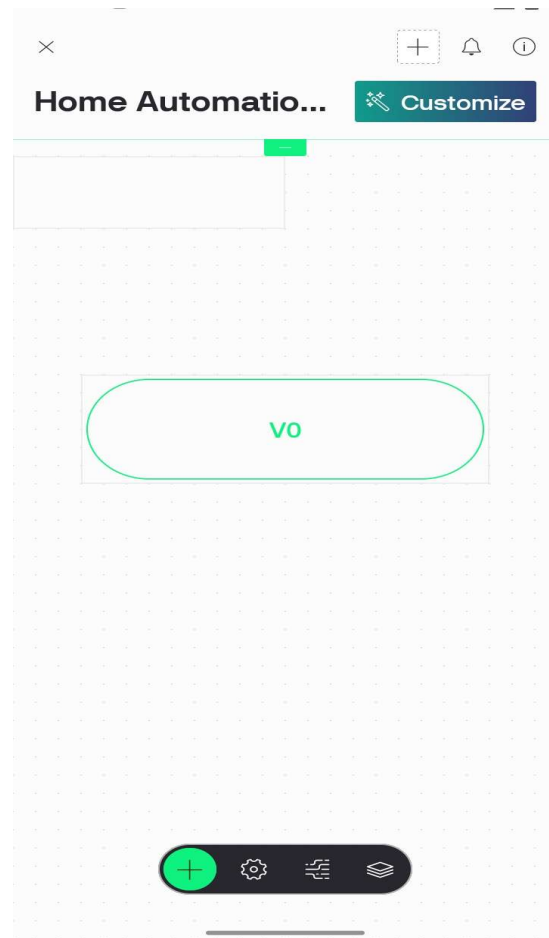
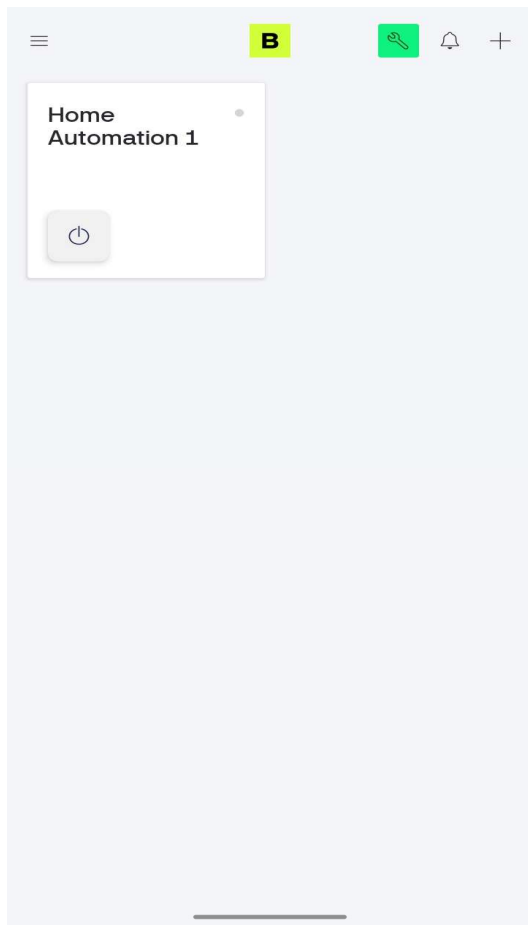
6.OUTPUT:

HARDWARE SETUP:



BLYNK APP SCREENSHOTS:





7.RESULT:

The smart home automation system successfully controls a relay-connected light through two synchronized input methods. The clap sensor accurately detects one and two consecutive claps within a 1-second window, turning the light OFF and ON respectively. The Blynk mobile application button provides simultaneous remote control over Wi-Fi, and both the relay state and the app button display remain synchronized at all times through the shared `applyState()` function. The use of rising-edge detection and debounce logic eliminates false triggering and ensures that each physical clap is counted exactly once. The system operates reliably without the use of `delay()` in the main loop, allowing uninterrupted Blynk communication throughout operation.

8.CONCLUSION:

The Smart Home Automation project demonstrates an effective and practical approach to controlling home appliances using both physical sensor input and remote IoT connectivity. By combining a clap sensor with the Blynk mobile application platform on an ESP32 microcontroller, the system achieves intuitive hands-free control alongside convenient app-based control. Key firmware techniques including rising-edge detection, time-window-based clap counting, and a centralized state management function ensure reliable, synchronized operation. This project serves as a scalable foundation for more advanced smart home systems that could include multiple appliances, voice control, scheduling, and energy monitoring. Overall, it highlights the potential of low-cost IoT solutions in improving everyday home convenience and safety.

9. REFERENCES:

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